

PCA

Predictive Maintenance

Industrial Motor Health Monitoring

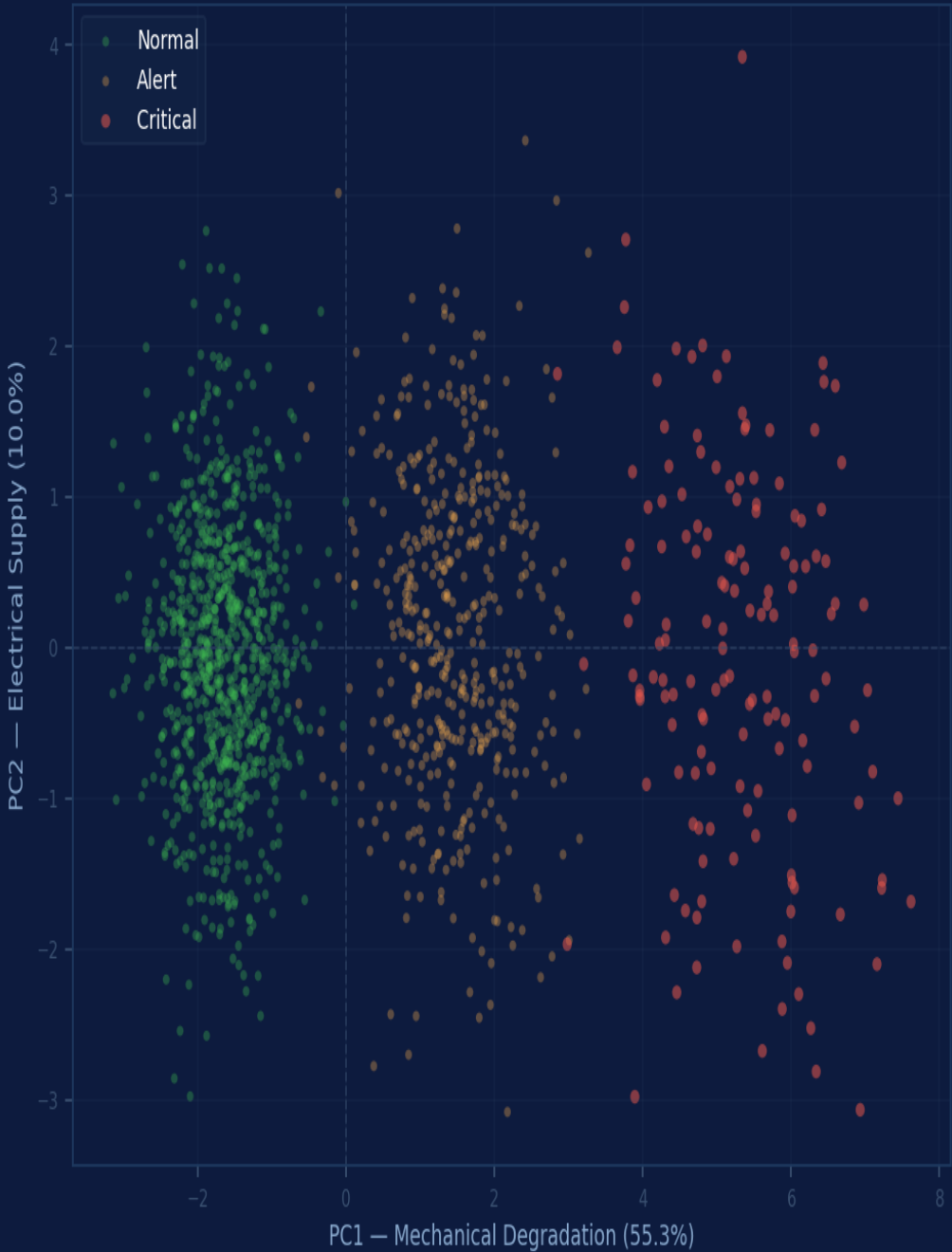
"Ten sensors produce noise.
Three PCs produce understanding."

PCA · k=3

StandardScaler

SPE · T²

Reliability Index



1,687

Motor Snapshots

10

Sensor Signals

54.4%

PC1 Variance

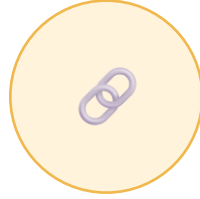
0–1

Reliability Index



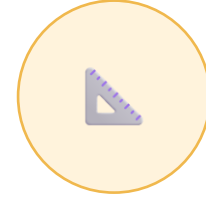
The Dashboard Problem

A maintenance engineer watches 10 trend charts. Seven look fine. Two show slight elevation. One is 12% below the alarm. He decides to wait. 72 hours later, the motor fails mid-shift. The information was there — distributed across all 10 channels simultaneously.



The Degradation Cascade

Bearing wear drives friction (vibration $\uparrow$). Friction drives heat (temperature $\uparrow$). Heat degrades lubrication (flow $\downarrow$, pressure $\downarrow$). Reduced lubrication increases load (current $\uparrow$). Load degrades insulation (power factor $\downarrow$). Ten sensors — one root cause.



The Geometry of Failure

From Normal to Critical: vibration RMS +210%, HF vibration +488%, bearing temp +44%, lube pressure -45%. These signals move together because they share the same physics. PCA finds the axis along which they move — and calls it PC1.

PCA compresses 10 correlated sensors into 3 orthogonal axes — and puts all the degradation signal in PC1.

10 TREND CHARTS

Vibration: 4.2 mm/s → below 5.0 alarm

Temperature: 78°C → within bounds

Lubrication: 2.8 bar → acceptable

→ *Each chart evaluated in isolation*

→ *No correlation between signals*

→ *Engineer sees 10 partial stories*

→ *Decision: wait for next inspection*

→ *Motor fails 72h later*



PCA HEALTH SPACE

All 10 sensors compressed into 3 PCs

PC1 captures 54.4% variance alone

PC1 = the mechanical degradation axis

→ *All correlated signals become one score*

→ *Reliability Index $IR \in (0, 1]$*

→ *Engineer sees one health trajectory*

→ *$IR = 0.71 \rightarrow$ Severe $\rightarrow$ schedule now*

→ *Motor inspected before failure*

SCADA

CbM System / Accelerometer

vibration_rms_mm_s · vibration_hf_g

Triaxial accelerometer at motor housing — RMS broadband and high-frequency band (bearing wear signature)

DCS

PLC / RTD

bearing_temp_c · speed_rpm

Resistance temperature detector at bearing housing · Proximity probe or encoder on motor shaft

PMR

Power Monitoring Relay

motor_current_a · voltage_v · power_factor_pct

CT-based phase current + voltage tap per phase — captures electrical load, supply quality, and insulation state

LUBE

Lubrication System / SCADA

lube_flow_l_min · lube_pressure_bar · internal_humidity_pct

Inline flow meter and pressure transducer on lube circuit · Capacitive humidity sensor inside motor enclosure

op_status (Normal / Alert / Critical) is used only for external validation — PCA never sees it during training.

"PC1 is not a statistical artifact. It is the degradation cascade that every motor engineer already knows exists — PCA gave it coordinates."

Continuous Score, Not Binary Label

DBSCAN flags anomalies. PCA tracks trajectories. A motor that is 20% of the way to failure is not anomalous — but it is actionable. The Reliability Index IR quantifies how far the motor has traveled from the healthy origin.

PC1 = Physical Interpretation

When loadings of vibration (+0.40), temperature (+0.39), current (+0.36) and lube pressure (−0.36) align into one axis, that axis has a name: mechanical degradation. PCA found it. The engineer understands it.

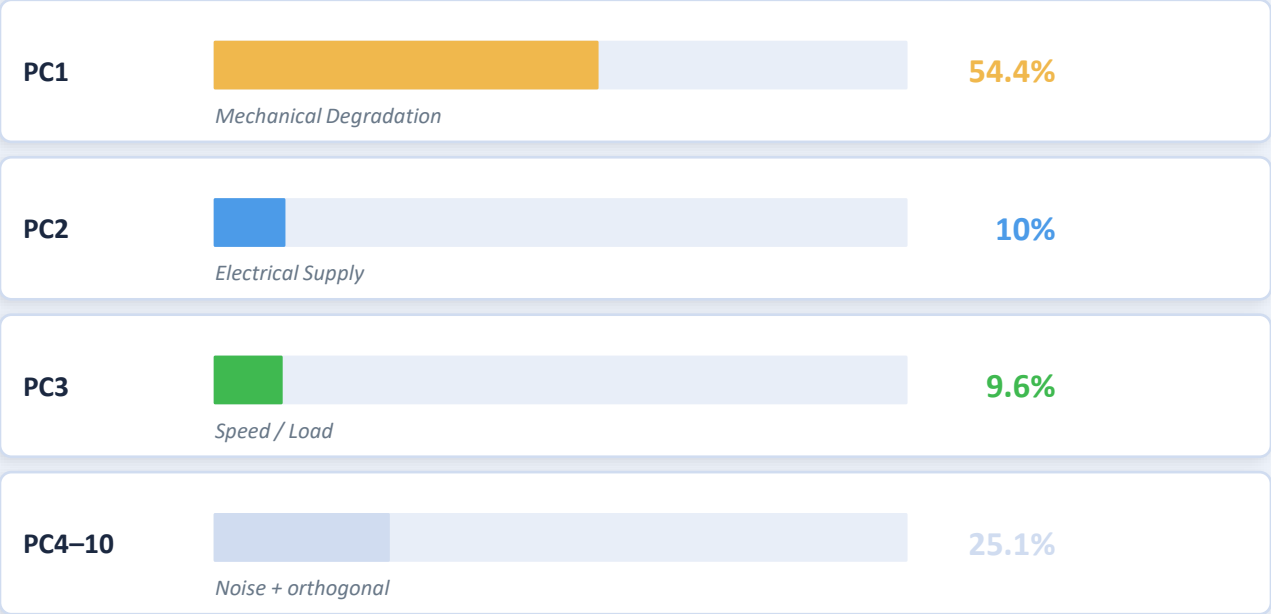
Separates Degradation from Operating Conditions

PC2 captures voltage (loading +0.97). PC3 captures speed (+0.92). These are orthogonal to PC1 — a motor can have supply voltage issues without being mechanically degraded. The health space separates these dimensions cleanly.

Tiered Early Warning

$IR \geq 0.86 \rightarrow$ Normal. $IR\ 0.74\text{--}0.86 \rightarrow$ Alert. $IR\ 0.62\text{--}0.74 \rightarrow$ Severe. $IR < 0.62 \rightarrow$ Critical. Four action tiers from one number — calibrated to the operating status ground truth without using it in training.

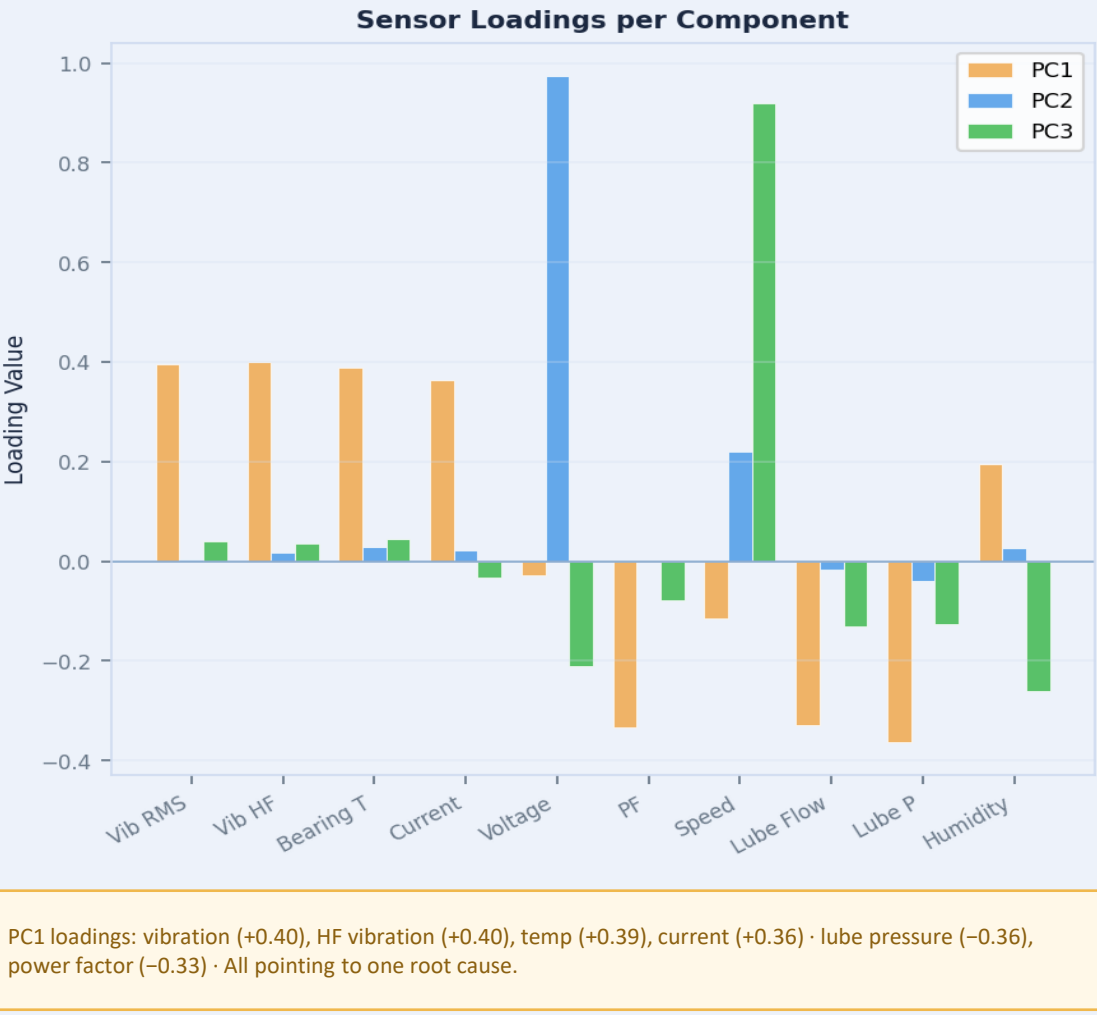
Variance & Health Index by Status



Reliability Index · Mean by Operating State



Sensor Loadings — What Each PC Represents



PC1 = 54.4% · 3 PCs = 74.0% · IR separates all three operating states without labels

Normal • IR ≥ 0.86

Trigger: SPE and T^2 within 95th percentile control limits • All 10 sensors within normal operating manifold



Continue production at full capacity. Log IR trend. Use as baseline for SPC monitoring. No maintenance action required.

Alert • IR 0.74 – 0.86

Trigger: T^2 or SPE approaching UCL • Vibration or temperature trending upward across multiple consecutive snapshots



Monitor closely. Visual inspection at next scheduled opportunity. Check lubrication flow and pressure manually. Increase monitoring frequency to every shift.

Severe • IR 0.62 – 0.74

Trigger: T^2 above UCL or SPE elevated • IR declining over 3+ consecutive readings — confirmed drift



Schedule inspection within 24–48 hours. Do not wait for the next planned maintenance window. Prepare bearing replacement kit and lubrication system components. Reduce load if possible.

Critical • IR < 0.62

Trigger: Both SPE and T^2 above UCL • IR below 0.62 confirmed on 2+ consecutive readings



Halt for immediate maintenance. Measure bearing clearance. Inspect lubrication system for blockage or pump failure. Replace bearing set. Do not restart until IR > 0.80 on post-maintenance verification.

A — Healthy Motor	
Vibration RMS	1.8 mm/s
HF Vibration	0.28 g
Bearing Temp	63°C
Current	16.4 A
Lube Pressure	3.8 bar
Power Factor	96%

 Normal • IR = 0.91

−2.1 (near origin)

B — Severe Degradation	
Vibration RMS	4.9 mm/s
HF Vibration	1.20 g
Bearing Temp	84°C
Current	24.1 A
Lube Pressure	2.6 bar
Power Factor	90%

 Severe • IR = 0.68

PC1 = +2.8 (drifting)

C — Critical · Stop Now	
Vibration RMS	7.1 mm/s
HF Vibration	2.40 g
Bearing Temp	98°C
Current	30.8 A
Lube Pressure	1.8 bar
Power Factor	84%

 Critical • IR = 0.54

PC1 = +5.4 (far from normal)

Motor B is not anomalous by any individual threshold — vibration 4.9 mm/s (limit: 5.0), temperature 84°C (limit: 95°C). PC1 = +2.8 and IR = 0.68 already classify it as Severe. Inspection scheduled 36h later. Bearing replaced before failure.

The map was always there.

**Drawn by 10 sensors,
read by 3 components.**

The degradation was not hiding. It was distributed — across vibration, temperature, current, lubrication. PCA found the axis along which all of them move together. That axis was the answer.

Download. Run it on your own motor historian data.

Change k . Add sensors. Recalibrate the IR thresholds for your process.

**Because every motor is already telling you what it needs.
You just need to listen to all 10 channels at once.**

